

Yafira Martinez

Design Engineer / Creative Technologist

interactive systems, interfaces, and experimental technologies

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EXPERIENCE

Teaching and Academic Support

Graduate Assistant — Intro to Wearables, NYU (IMA)

Jan 2026 – Present

Adjunct Instructor: Daniel Ryan Johnston

- Supporting instruction in wearable and e-textile design, focusing on body-centered interaction, gesture, and non-verbal communication.
- Assisting students with hands-on projects with soft circuits and Adafruit Gemma M0, using Arduino code with light and motion sensors to prototype expressive wearables.

Graduate Assistant — Creative Computing, NYU (IMA)

Sept 2025 – Dec 2025

Professor: Daniel Shiffman

- Supported instruction and mentored students in creative coding through office hours, workshops, and assignment feedback.

Research Assistant — Computational Text, NYU (ITP/IMA)

May 2025–Aug 2025

Professor: Daniel Shiffman

- Assisted in developing the Computational Text course syllabus by identifying tools, artist projects, and frameworks in procedural text, data ethics, and machine learning.
- Researched and curated readings and case studies bridging procedural text, data ethics, and machine learning for creative coding contexts.

Research & Open Source

Machine Learning Researcher & Web Contributor — ml5.js

June 2025 – Present

- Developing ml5.transcribe() and ml5.embeddings(), expanding access to browser-based transformer models (Whisper, Moonshine) for creative coding.
- Refining interaction design, inference performance, and web tooling to support real-time creative ML and contributor usability.

Accessibility Researcher — Making Fashion Accessible, Vertically Integrated Projects (VIP), NYU Ability Lab

Oct 2025–Present

- Researching accessible fashion technologies using knitting machines and AYAB to support inclusive, open-source garment production.
- Contributing to adaptable textile workflows centered on accessibility and human-centered design.

Development & Community

Lead Web Developer — Equitable Syllabus Project, NYU (ITP/IMA)

Nov 2024 – Present

- Designing and implementing accessible, user-centered interfaces using React.js and Figma.
- Integrating and migrating project data while maintaining centralized contribution tracking (Google Sheets) to improve usability and sustainability.

Web Developer & Designer — Useless Machines Archive

Mar 2025 – Aug 2025

Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

Camp Counselor — ITP Camp, New York University

June 2025 - July 2025

- Supported a 150+ person creative technology community through workshops, project mentorship, and hands-on troubleshooting across hardware, software, and fabrication during a two-week intensive.

DESIGN

UX/UI Design, Interaction Design, Responsive Design, Prototyping, Wireframing, Design Systems, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite

SOFTWARE

JavaScript, HTML/CSS, React, Next.js, Node.js, APIs/REST APIs, Web Serial, p5.js, ml5.js, Python, CircuitPython, Creative Coding, Git

HARDWARE

Physical Computing, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication, Microcontrollers, Arduino IDE, Raspberry Pi, Adafruit, Fritzing, Laser Cutter, Knitting Machine

CERTIFICATIONS

Artificial Intelligence,

Samsung Innovation Campus
November 2022 – May 2023

Human Computer Interaction,

Interaction Design Foundation
February 2022

Google UX Design, Coursera

July 2021 – December 2021

EDUCATION

Master of Professional Studies,
Interactive Telecommunications
Program (ITP)

New York University –
Tisch School of Arts
Expected: May 2026

Bachelor of Science,
Computer Science

City University of New York -
Lehman College
December 2020

SELECTED WORK & FEATURES

Soft Circuit Library — Electronics Faire (Theme: Repair), Temple University Libraries

April 2026

- Exhibited a collection of handmade textile circuit boards organized as a living archive, each designed to be handled, studied, and repaired by touch. Visible repair notes and patched connections treat maintenance as part of the circuit's ongoing life.

Guardians of the Groceries (G.O.T.G.), — PlayfulNYC Hackathon 2026, Parsons School of Design

April 2026

- Built the hardware and software layer of a handheld ESP32-based device that pairs with a live web registry to document NYC bodega cats. It captures a photo, uploads it to Supabase via WiFi, and logs each cat's alter ego, hidden powers, and cuteness rating. **Awarded** ☆ **Best of Software** ☆

Open Hardware Summit Fellow — Open Source Hardware Association, Berlin, Germany

Feb 2026 - present

- Selected fellow exhibiting NYU ITP thesis "The Soft Computer" and other experimental open-source hardware projects.

Pocket Oracle of Time — Horological Society of New York (HSNY)

Feb 2026 - April 2026

- Exhibited a poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.

MIT Hard Mode Hackathon, AI x Hardware 2026 — MIT Media Lab

Mar 2026

- Built *enoki*, a focus companion for communal accountability — a React Native app using Mentra smart glasses and Claude Vision AI paired with a desktop companion device using Raspberry Pi, Seed sensors, and motors, both designed to work together and visualize focus as a mushroom that grows when on task and wilts when distracted.

Moon Pocket — Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor

Jan 2026

- Showcased a connected device built with an Arduino Nano 33 IoT that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.

Tendertronics: Analog Circuit Performance — NYU ITP's NIME Show, Littlefield

Dec 2025

- Performed an analog instrument exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.

Lightning Talk Speaker — Exploring Creativity in the Age of AI, NYU Global Show & Tell

Oct 2025

- Presented *Open Tools, Soft Interfaces*, discussing contributions to ml5.js and open-source machine learning, and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.

UN Open Source Week Hackathon, United Nations Headquarters

June 2025

- Participated in a global hackathon focused on open-source solutions for child-centered climate emergency response; contributed to the Ahead of the Storm challenge using geospatial data.

The Internet Book — Annual publication on the poetic web

May 2025

- Featured for electrocute.io, a collection of soft-tech and poetic computing experiments, in a curated directory of designers, developers, writers, and digital artists.

Static Chic — Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor

May 2025

- Presented *Static Chic*, a 3D-printed inductive handbag that wirelessly lights up embedded LEDs. Part of a wearables and e-textiles open mic curated by Kari Love and Liza Stark.

Soft Glitch — eTextile Swatch, Radiant Fibers Exhibition, NYU Clive Davis Gallery

Jan 2025 - Mar 2025

- Exhibited an e-textile swatch combining punch needle, chaotic yarns, and electroluminescent thread to explore digital disruption through texture and light.

MIT Reality Hack 2025 — Massachusetts Institute of Technology

Jan 2025

- Implemented sound and haptic feedback for *HangarPro VR*, a virtual reality training simulation using Cognitive3D to enhance pre-flight safety through visual interaction analysis for aerospace exploration.